Pneumonic plague is a lung infection caused by the bacterium Yersinia pestis. If left untreated, it is nearly 100% fatal.

Pneumonic plague can spread person-to-person through respiratory droplets with close and sustained contact.

Cats can also spread pneumonic plague to humans, if the plague bacteria (Yersinia pestis) has infected the cat's lungs.

The symptoms of pneumonic plague are Fever, Shortness of breath, Chest pain, Cough, Sputum (thick mucus produced by the lungs)

The incubation period of pneumonic plague can be as short as one day after exposure.

All forms of the plague are caused by the bacterium Yersinia pestis. In addition to pneumonic plague, this also includes:

Bubonic plague: The most common form of plague and the plague associated with the Black Death. It is caused by an infected flea bite, causing painful, swollen lymph nodes, often filled with pus.

Septicemic plague: A severe form of plague in which the bacteria spreads through the bloodstream. It can occur on its own or develop from bubonic plague.

Pharyngeal plague: A less common form of plague that affects the throat. This occurs after eating uncooked or undercooked meat from an infected animal.

If bubonic plague or septicemic plague are left untreated, they can spread to the lungs as pneumonic plague.

How is pneumonic plague diagnosed?

Plague can be identified by Blood test and Cultures of respiratory samples (such as sputum or lung fluid)

Lab staff need to be alerted anytime plague is suspected so they can protect themselves by wearing protective gear.

For those who have been exposed to pneumonic plague, antibiotics are the best treatment option to prevent active infection. Doxycycline, one of the tetracycline drugs, or ciprofloxacin, an antibiotic from the fluoroquinolone drug class, are often prescribed. Recovery rates are high if treated quickly – less than 24 hours after symptoms begin.

For those who have pneumonic plague, the treatment would be IV antibiotics, typically old school antibiotics like gentamicin. But also fluoroquinolones would be options in patients who can't tolerate the aminoglycosides.

The transmission rate (R0) for pneumonic plague is about 1.3. It means that someone with pneumonic plague would infect an average of 1.3 people. By comparison, measles has an R0 of 12-18.